

Older Than They Used to Be

84-355 Democracy's Data | The age of government actors, and the denominator that decides the answer

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Everything here runs as given. **You will not write any code.** Your job is to read what comes out and say what it means.

Read `officeholder-age-brief.Rmd` first. This lab runs the same steps with the code exposed.

Where did that number come from?

"Congress is older than it has ever been."

"Sure, but people live thirty years longer than they used to."

The first sentence is checkable. The second sounds like it settles the matter and does not. This lab is about why.

Step 1 — One file, and what a birth *year* buys you

```
m <- read.csv(file.path(D, "members.csv"), stringsAsFactors = FALSE)
ac <- read.csv(file.path(D, "age_by_congress.csv"), stringsAsFactors = FALSE)
le <- read.csv(file.path(D, "life_expectancy.csv"), stringsAsFactors = FALSE)
et <- read.csv(file.path(D, "entry_tenure.csv"), stringsAsFactors = FALSE)
pr <- read.csv(file.path(D, "presidents.csv"), stringsAsFactors = FALSE)

c(rows      = nrow(m),
  congresses = length(unique(m$congress)),
  first_year = min(m$year),
  last_year  = max(m$year),
  no_birth_yr = sum(is.na(m$born)))
```

```
#>      rows congresses first_year last_year no_birth_yr
#>    51062         119      1789     2025         176
```

```
table(m$chamber)
```

```
#>
#>      House President      Senate
#>    40933         129    10000
```

Every age here comes from one column, **born**, in the Voteview member file, and **born** holds a **year**, not a **date**. Age in this lab means *the year the Congress convened minus the year the member was born*.

```
pr[pr$bioname == "BIDEN, Joseph Robinette (Joe), Jr.",
  c("bioname", "born", "year", "age_at_term_start")]
```

```
#>
#>      bioname born year age_at_term_start
#> 46 BIDEN, Joseph Robinette (Joe), Jr. 1942 2021          79
```

Joe Biden was inaugurated in January 2021 aged **78**; his birthday is in November. Year arithmetic says 79.

Every age in this lab is approximate to within a year, and the errors point in both directions, so medians and trends survive it and individual ages do not.

```
c(rows_with_no_birth_year = sum(is.na(m$born)),
  of_those_before_1830    = sum(is.na(m$born) & m$year < 1830))
```

```
#> rows_with_no_birth_year    of_those_before_1830
#>                        176                        127
```

```
# one person, some rows with a birth year and some without
sp <- split(m$born, m$icpsr)
unique(m$bioname[m$icpsr %in%
  as.integer(names(sp)[sapply(sp, function(b) any(is.na(b)) & any(!is.na(b))])])])
```

```
#> [1] "CLEVELAND, Stephen Grover"
```

176 rows have no birth year at all, almost all of them in the first twenty Congresses. **And exactly one person has a birth year on some of his rows and not on others**: Grover Cleveland, recorded on his first term and blank on his second. Nothing in the file announces that. An average would skip the blank rows and hand back a number that looked perfectly healthy.

Step 2 — How old is Congress, and when did it change?

```
C <- ac[ac$chamber == "Congress", ]
c(youngest      = min(C$median_age),
  youngest_year = C$year[which.min(C$median_age)],
  oldest        = max(C$median_age),
  now           = C$median_age[C$congress == max(C$congress)],
  senate_now    = ac$median_age[ac$chamber == "Senate" &
                                ac$congress == max(ac$congress)],
  house_now     = ac$median_age[ac$chamber == "House" &
                                ac$congress == max(ac$congress)])
```

```
#>      youngest youngest_year      oldest      now      senate_now
#>      42          1793          60          59          65
#>      house_now
#>      58
```

```
C$year[C$median_age == max(C$median_age)]      # every Congress at the max
```

```
#> [1] 2017 2019 2021
```

```
TR      <- C[C$year >= 1965 & C$year <= 1995, ]
TR      <- TR[which.min(TR$median_age), ]
PRIOR   <- max(C$year[C$year < TR$year & C$median_age <= TR$median_age])
c(trough_year = TR$year, trough_median = TR$median_age,
  last_time_that_young = PRIOR)
```

```
#>      trough_year      trough_median last_time_that_young
#>      1979          50          1913
```

The rise is real and it is recent. The median member was 50 in 1979 — a figure Congress had not touched since 1913 — and 60 by 2017. Almost all of the increase people argue about happened after 1980.

The Senate has always been the older chamber: 65 against 58 today.

```
c(n_presidencies = nrow(pr),
  presidencies_with_no_age = sum(is.na(pr$age_at_term_start)),
  mean_before_1900 = round(mean(pr$age_at_term_start[pr$year < 1900], na.rm = TRUE), 1),
  mean_since_1981 = round(mean(pr$age_at_term_start[pr$year >= 1981], na.rm = TRUE), 1))
```

```
#>      n_presidencies presidencies_with_no_age      mean_before_1900
#>      47.0          5.0          55.0
#>      mean_since_1981
#>      64.2
```

```
head(pr[order(-pr$age_at_term_start), c("bioname", "year", "age_at_term_start")], 4)
```

```
#>      bioname year age_at_term_start
#> 46 BIDEN, Joseph Robinette (Joe), Jr. 2021      79
#> 47      TRUMP, Donald John 2025      79
#> 45      TRUMP, Donald John 2017      71
#> 40      REAGAN, Ronald Wilson 1981      70
```

47 presidencies is not enough observations to support a trend line, and this lab does not draw one. The arithmetic is still worth having: the average age on taking office was 55.0 before 1900 and 64.2 since 1981.

Step 3 — The rebuttal, stated as strongly as it deserves

```

sitting <- function(y) floor((y - 1787) / 2) # the Congress sitting in year y
BY <- 1950; EY <- 2018                     # the window where both life
                                             # expectancy series exist

A0 <- C$median_age[C$congress == sitting(BY)]
A1 <- C$median_age[C$congress == sitting(EY)]
E0a <- le$e0_hus[le$year == BY]; E0b <- le$e0_hus[le$year == EY]

c(congress_median_1950 = A0, congress_median_2018 = A1, gain = A1 - A0,
  life_expect_1950 = E0a, life_expect_2018 = E0b, gain_le = E0b - E0a)

```

```

#> congress_median_1950 congress_median_2018      gain
#>                53.0                60.0         7.0
#>    life_expect_1950    life_expect_2018    gain_le
#>                68.2                78.7        10.5

```

Life expectancy rose **10.5 years** while the median member of Congress aged **7**. On that comparison Congress has aged *less* than the country it is drawn from, and the rebuttal wins.

That comparison is wrong.

Step 4 — The denominator that flips the answer

78.7 is life expectancy **at birth** — the average length of a life for a baby born in 2018. It rose so much across the twentieth century mostly because **babies and children stopped dying**, and a sitting senator has already survived all of that.

The number that describes an officeholder is **remaining life expectancy at an age they have reached**. NCHS publishes it in the same table, one section down the page.

```

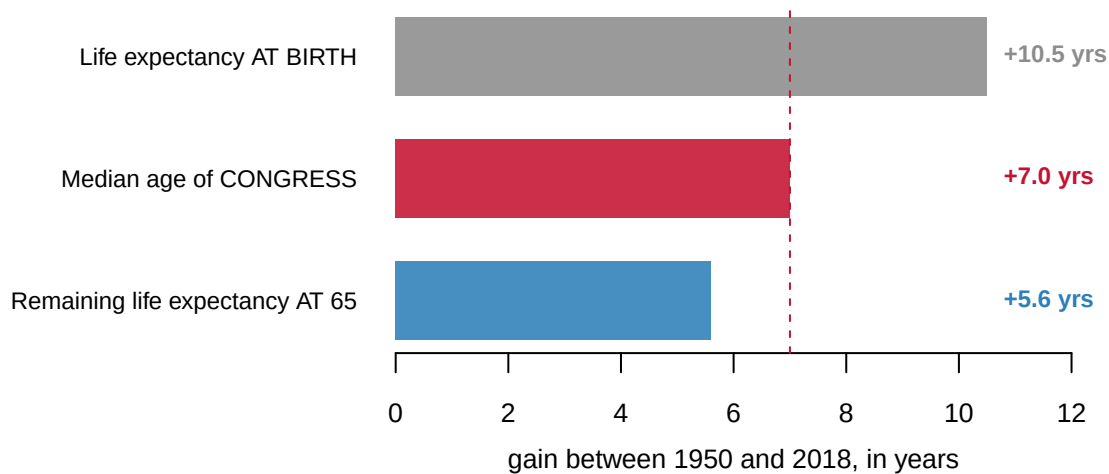
E6a <- le$e65[le$year == BY]; E6b <- le$e65[le$year == EY]
c(remaining_at_65_1950 = E6a, remaining_at_65_2018 = E6b, gain_e65 = E6b - E6a,
  congress_gain = A1 - A0,
  share_of_headline_gain_after_65 = round(100 * (E6b - E6a) / (E0b - E0a)))

```

```

#>          remaining_at_65_1950    remaining_at_65_2018
#>                13.9                19.5
#>          gain_e65          congress_gain
#>                5.6                7.0
#> share_of_headline_gain_after_65
#>                53.0

```



Congress aged 7 years. Remaining life expectancy at 65 grew 5.6. Against the at-birth yardstick Congress aged slower than the country; against the honest one it aged faster. Same source, same table, different row.

At most 53% of the famous rise in life expectancy is a rise in how long a 65-year-old lives. The rest is people who used to die young and now do not.

A comparison has a denominator, the denominator is a choice, and a denominator chosen without thinking can reverse the sign of the answer.

What the at-65 series will not do

```
c(years_with_e65 = sum(!is.na(le$e65)),
  first_year      = min(le$year[!is.na(le$e65)]),
  last_year       = max(le$year[!is.na(le$e65)]),
  years_with_e0   = sum(!is.na(le$e0_annual)))
```

```
#> years_with_e65    first_year    last_year  years_with_e0
#>              25         1950         2018         119
```

```
le$year[!is.na(le$e65)]
```

```
#> [1] 1950 1960 1970 1975 1980 1990 1995 2000 2001 2002 2003 2004 2005 2006 2007
#> [16] 2008 2009 2010 2012 2013 2014 2015 2016 2017 2018
```

The at-birth series is annual back to 1900. **The at-65 series exists for 25 years only**, none of them before 1950. Nothing between the published years has been interpolated here. The comparison in this step can be made for 1950 onward and **cannot honestly be made for 1900** — which is a pity, because 1900 is where the at-birth number looks most dramatic.

The two NCHS publications of life expectancy at birth also disagree with each other:

```
ov <- le[!is.na(le$e0_hus) & !is.na(le$e0_annual), ]
c(shared_years = nrow(ov),
  agree_exactly = sum(ov$e0_hus == ov$e0_annual),
  worst_gap     = max(abs(ov$e0_hus - ov$e0_annual)))
```

```
#> shared_years agree_exactly worst_gap
#>          26.0          24.0          0.4
```

```
ov[ov$e0_hus != ov$e0_annual, c("year", "e0_annual", "e0_hus")]
```

```
#>      year e0_annual e0_hus
#> 104 2003      77.6   77.2
#> 105 2004      77.5   77.6
```

On 24 of 26 shared years they agree to the tenth. On two they do not, and the pattern — one series monotone, the other not — is what a transposed pair looks like. **This lab does not adjudicate it. It reports it.**

Step 5 — Entering older, or staying longer?

A rising median age has exactly two possible sources: **new members arrive older**, or **sitting members stay longer**. For every member, at every Congress, age now = age on arriving + years since. That is an identity, so it survives averaging.

```
dec <- function(ch, y1, y2) {
  a <- ac[ac$chamber == ch & ac$congress == sitting(y1), ]
  b <- ac[ac$chamber == ch & ac$congress == sitting(y2), ]
  round(c(rise_in_mean_age = b$mean_age - a$mean_age,
          from_arriving_older = b$mean_entry_age - a$mean_entry_age,
          from_staying_longer = b$mean_years_since_entry -
                                a$mean_years_since_entry), 2)
}
rbind(Congress = dec("Congress", BY, EY),
      House     = dec("House",    BY, EY),
      Senate    = dec("Senate",   BY, EY))
```

```
#>      rise_in_mean_age from_arriving_older from_staying_longer
#> Congress          6.10              3.86              2.25
#> House             6.28              4.25              2.03
#> Senate             5.55              2.40              3.15
```

```
DC <- dec("Congress", BY, EY); DH <- dec("House", BY, EY); DS <- dec("Senate", BY, EY)
round(c(Congress = 100 * DC[2] / DC[1],
      House     = 100 * DH[2] / DH[1],
      Senate    = 100 * DS[2] / DS[1]))
```

```
#> Congress.from_arriving_older House.from_arriving_older
#>              63              68
#> Senate.from_arriving_older
#>              43
```

63% of the rise is people arriving older, not incumbents hanging on. And the chambers disagree: it is 68% in the House and 43% in the Senate, so **most of the Senate's ageing really is senators staying** — which six-year terms and very high re-election rates would predict.

```

rbind(`1950 to 1980` = dec("Congress", 1950, 1980),
      `1980 to 2025` = dec("Congress", 1980, 2025))

```

```

#>           rise_in_mean_age from_arriving_older from_staying_longer
#> 1950 to 1980           -2.68           -3.50           0.82
#> 1980 to 2025           8.71           7.45           1.26

```

Congress got **younger** through the 1960s and 1970s, and did it by electing younger people: arrival age fell 3.5 years while time in office barely moved. Then the same lever went the other way.

Step 6 — The median hides the tail

```

q <- c("pct_65plus", "pct_70plus", "pct_80plus")
round(rbind(
  `Congress 1950` = unlist(ac[ac$chamber == "Congress" & ac$congress == sitting(BY), q]),
  `Congress 2025` = unlist(ac[ac$chamber == "Congress" & ac$congress == max(ac$congress), q]),
  `Senate 2025`   = unlist(ac[ac$chamber == "Senate"   & ac$congress == max(ac$congress), q])), 1)

```

```

#>           pct_65plus pct_70plus pct_80plus
#> Congress 1950       16.1       8.5       1.1
#> Congress 2025       34.9      24.1       4.4
#> Senate 2025        53.4      35.9       5.8

```

```

hs <- m[m$chamber %in% c("House", "Senate") & !is.na(m$age), ]
Q <- function(y) quantile(hs$age[hs$congress == sitting(y)], c(0.05, 0.5, 0.95))
round(rbind(`1950` = Q(BY), `2025` = Q(2025), `moved by` = Q(2025) - Q(BY)))

```

```

#>           5% 50% 95%
#> 1950       36  53  72
#> 2025       39  59  79
#> moved by   3   6   7

```

The middle moved; the top moved further. The share of Congress over 70 went from 8.5% to 24.1%, and in the Senate it is now 35.9%. Between 1950 and 2025 the 5th percentile of age rose 3 years, the median 6, and the 95th 7. **The distribution did not slide. It stretched, at the top.**

Step 7 — Careers, and the censoring trap

```

E2 <- et[!et$still_serving & !is.na(et$entry_age) & et$entry_year <= 1985, ]
E2$cohort <- cut(E2$entry_year, c(1789, 1849, 1909, 1949, 1986), right = FALSE,
  labels = c("1789-1848", "1849-1908", "1909-1948", "1949-1985"))
do.call(rbind, lapply(levels(E2$cohort), function(g) {
  z <- E2[E2$cohort == g, ]
  data.frame(entered = g, careers = nrow(z),
    median_entry_age = median(z$entry_age),
    median_years_served = median(z$years_served),
    pct_20yrs_plus = round(100 * mean(z$years_served >= 20), 1))
}))

```

```
#>      entered careers median_entry_age median_years_served pct_20yrs_plus
#> 1 1789-1848    2901             41             4             1.1
#> 2 1849-1908    4392             45             4             3.7
#> 3 1909-1948    2578             48             6            12.1
#> 4 1949-1985    1688             44            10            21.1
```

Notice what that table excluded: everybody still serving, and everybody who entered after 1985.

```
c(careers_in_file = nrow(et),
  still_serving   = sum(et$still_serving),
  entered_since_1986 = sum(et$entry_year >= 1986))
```

```
#>      careers_in_file      still_serving entered_since_1986
#>                13320                554             1668
```

A career you can measure is a career that is over. Ask “how long do members serve?” of everyone in the file and the recent cohorts look short — not because service got shorter but because it has not finished. That is **right-censoring**, and it is the commonest way a tenure statistic gets reported backwards.

```
S <- et[et$chamber == "Senate" & !is.na(et$entry_age), ]
c(pct_with_prior_house_all_time = round(100 * mean(S$prior_house), 1),
  pct_with_prior_house_since_1981 =
    round(100 * mean(S$prior_house[S$entry_year >= 1981]), 1),
  mean_entry_age_with_house =
    round(mean(S$entry_age[S$entry_year >= 1981 & S$prior_house]), 1),
  mean_entry_age_without =
    round(mean(S$entry_age[S$entry_year >= 1981 & !S$prior_house]), 1))
```

```
#>      pct_with_prior_house_all_time pct_with_prior_house_since_1981
#>                31.7                42.1
#>      mean_entry_age_with_house      mean_entry_age_without
#>                51.6                53.3
```

A popular explanation for senators arriving old is that they come up through the House. They increasingly do — 42.1% of those entering since 1981 — but **those senators arrive younger**, not older (51.6 against 53.3). The pipeline is not the cause.

Step 8 — Who leaves the data before you can count them

```
died_in_office <- !is.na(hs$died) & hs$died >= hs$year & hs$died <= hs$year + 1
era <- ifelse(hs$year < 1900, "before 1900",
  ifelse(hs$year < 1950, "1900-1949",
    ifelse(hs$year < 2000, "1950-1999", "2000 and after")))
round(100 * tapply(died_in_office, factor(era, c("before 1900", "1900-1949",
  "1950-1999", "2000 and after")),
  mean), 2)
```

```
#>      before 1900      1900-1949      1950-1999 2000 and after
#>                2.18                3.20                1.50                0.57
```

```
c(member_terms = length(died_in_office), ended_in_death = sum(died_in_office))
```

```
#>   member_terms ended_in_death  
#>           50768           1038
```

Members used to die in office several times as often as they do now. **Each of those deaths removed an old member from every later Congress in this file**, so the historical series is biased downward — in exactly the direction that makes Congress look as though it used to prefer the young.

That bias *is* explained by longevity, and it belongs in the same argument as Step 4, on the other side of the ledger.

Step 9 — What you can now say

You can say that Congress is at or near its oldest recorded point, and that the rise is recent rather than gradual.

You can say that “people just live longer” does not account for it, and how much of it that argument could account for at most: 5.6 years against a 7-year rise.

You can say which mechanism dominates, and that it differs by chamber: 68% arrival age in the House, 43% in the Senate.

You cannot say why. Nothing in this file identifies a cause.

You cannot say it matters. No measure of performance appears anywhere here.

Your turn

Change **one** thing, re-run, and write about what changed. Pick one.

Option A — Use the wrong denominator on purpose. Redo Step 4 with `e75` instead of `e65`, and then with `e0_annual`. Report all three answers and say which one you would put in a news story, and why. There is a defensible case for more than one.

Option B — Pick a different window. Step 4 uses 1950 to 2018 because that is where both series exist. Redo it 1950-1980 and 1980-2018 separately. Does the flip survive in both halves? If it does not, say what that means for the headline claim.

Option C — One state, or one party. `members.csv` carries `state_abbrev` and `party_code` (100 = Democrat, 200 = Republican). Compute median age by party over time, or for the delegation of a state you care about. Small groups are noisy; say something about how noisy before you interpret a gap.

Option D — The decomposition, one chamber at a time. Step 5 shows the House and the Senate behaving differently. Find the Congress where they diverged and say what was happening in American politics around then. The data will not tell you; you have to bring that.

Option E — Your own question. 51,062 member-Congress records, every Congress since 1789.

What this lab cannot tell you

- **Why members arrive older or stay longer.** Primaries, money, safe seats, party recruitment, longer pre-political careers and the absence of any retirement rule are all consistent with these numbers. Nothing here separates them.
- **Whether age affects how anyone governs.** No performance measure appears in this file, and none of the figures should be read as if one did.
- **Anything about who dies in office,** beyond the raw rate. Members who died are missing from every later Congress, which flattens the historical series (Step 8), and this lab measures the size of that effect but cannot remove it.
- **Anything about candidates who lost.** Every person in this file won. Whether the electorate is choosing older candidates or being offered older candidates is invisible here, and it is the most important thing the data does not carry.
- **A single person's exact age.** born is a year (Step 1).

On the discussion board

By Friday, 11:59 p.m.

- Step 4 changes one row of one table and reverses the conclusion. Write the sentence a journalist would honestly write about Congress and longevity, in one sentence, using both numbers.
- Step 5 says most of the rise is people arriving older, not incumbents staying. What would you expect to see in the data if the opposite were true?
- Step 8 shows that members used to die in office much more often. Does that make the historical comparison in Step 2 better or worse, and in which direction?

Where the data came from

- **Members of Congress and presidents, 1789 to present** — Voteview, *Congressional Roll-Call Votes Database* (Lewis, Poole, Rosenthal, Boche, Rudkin and Sonnet). https://voteview.com/static/data/out/members/HSall_members.csv One row per member per Congress, with **born** and **died** as years. The same file the **dw-nominate** lab uses for something completely different, which is worth noticing: one well-built file answers many questions.
- **Life expectancy at birth, 1900-2018, annual** — National Center for Health Statistics, via CDC open data. <https://data.cdc.gov/api/views/w9j2-ggv5/rows.csv?accessType=DOWNLOAD>
- **Remaining life expectancy at 65 and at 75, selected years 1950-2018** — National Center for Health Statistics, *Health, United States, 2019*, Trend Table 4. <https://www.cdc.gov/nchs/data/hus/2019/004-508.pdf> Parsed from the published PDF, because this series is not in CDC's open-data catalogue, and checked against the at-birth series above.

The build is `data/build-data.R`.